PAPER READING

Neural Networks Are Already Clean

***Link**: https://already-clean.goodfire.com/*

21 / 10 / 2026 · Jingshen Zhang

A Queston

Do you think the neural network can be interpreted, or not?

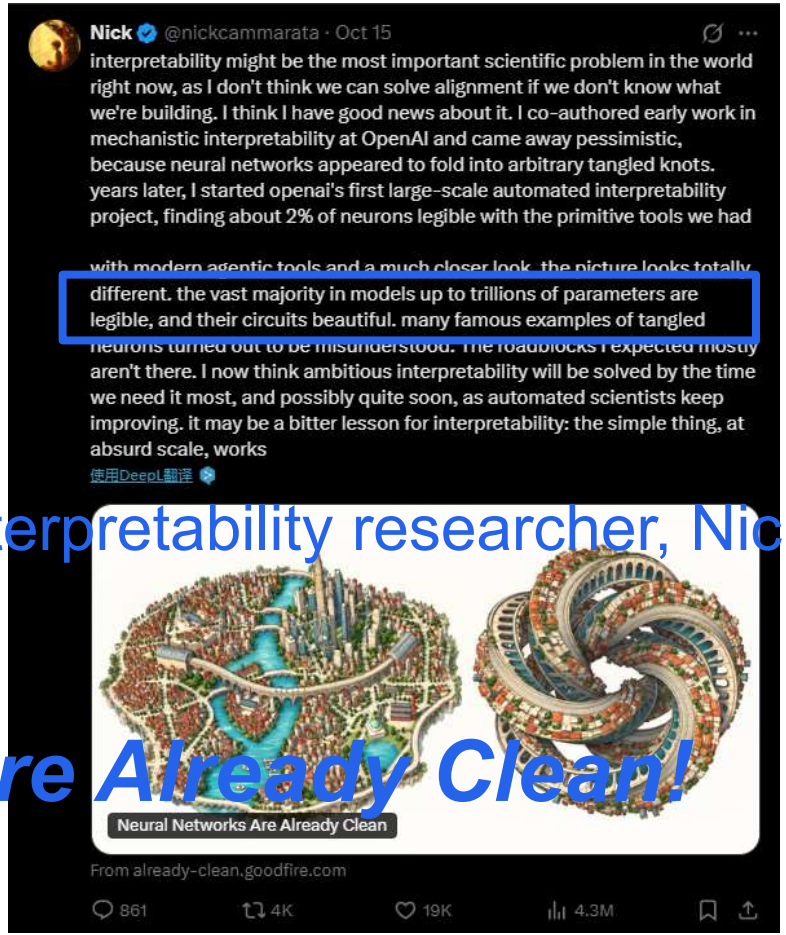
Background

- interpretability might be **the most important scientific problem** in the world right now, we can not solve alignment if we don't know what we're building.
- Unfortunately, the dominant view has long been that neural networks are heavily shaped by **superposition**, resulting in tangled representations that are difficult to interpret.

Good News

The viral X post from an interpretability researcher, Nick, brought good news:

Neural Networks Are Already Clean!



Outline

1. **Revisiting Canonical(classic) Evidence**
   - Polysemantic Circuits and Superposition
2. **Neural Networks Are Already Clean**
   - Lessons from Vision Models (why focus on vision? easier to manage, visualize, and analyze.)
3. **Discussion and Open Questions**

1. Revisiting Canonical Evidence

Two ways to approach mech. interpretability

1. Rapid normal science on the network itself.

2. A List of hypothesized features and tries to extract them all, e.g., SAE.

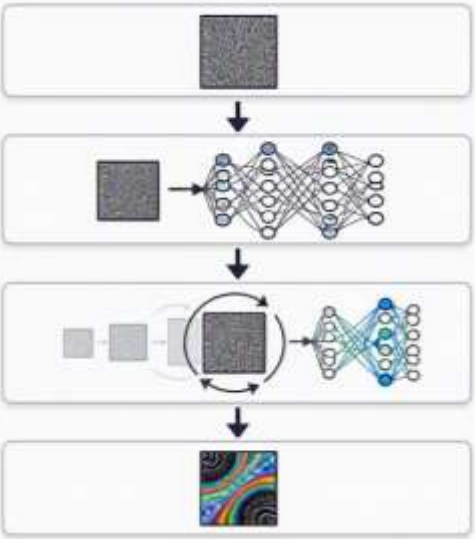
The field began with the first and fell back to the second because neural networks seemed to be tangled knots.

Interpret the original network vs. Reconstruct before interpreting

1. Revisiting Canonical Evidence

Feature Visualization → Cat-and-Car Neurons

- Given a neuron, start from a randomly initialized image.
- Pass the image through network and record the target neuron's activation.
- Perform gradient ascent on the input image to increase the activation, i.e., maximizing activation.
- Repeat until convergence and visualize the resulting image.



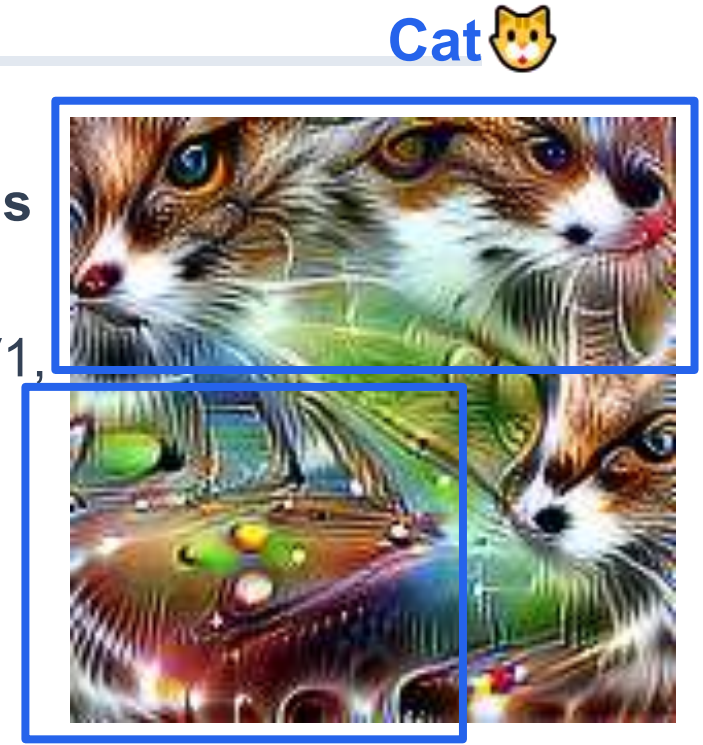
https://distill.pub/2017/feature-visualization/

1. Revisiting Canonical Evidence

Feature Visualization → Cat-and-Car Neurons

→ The famous Cat-and-Car neuron (InceptionV1, mixed4e:55) responded strongly to both cat faces and car fronts.

“isn’t responding to some commonality of cars and cat faces.”

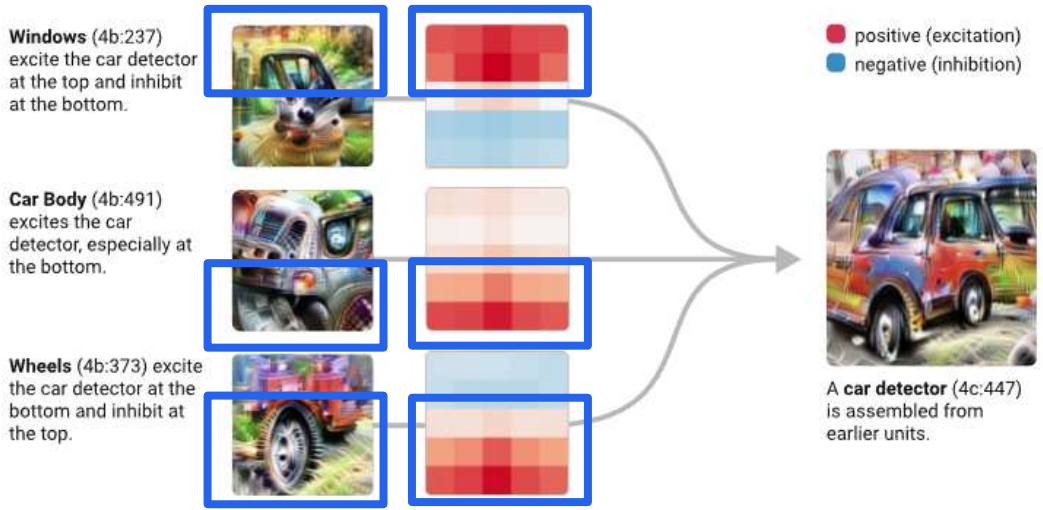


https://distill.pub/2017/feature-visualization/

1. Revisiting Canonical Evidence

Circuit → Dog-and-Car Circuit

- In mixed4c, a mid-late layer of InceptionV1, there is a car detecting neuron. Using features from the previous layers, it looks for wheels at the bottom of its convolutional window, and windows at the top.



https://distill.pub/2020/circuits/zoom-in/

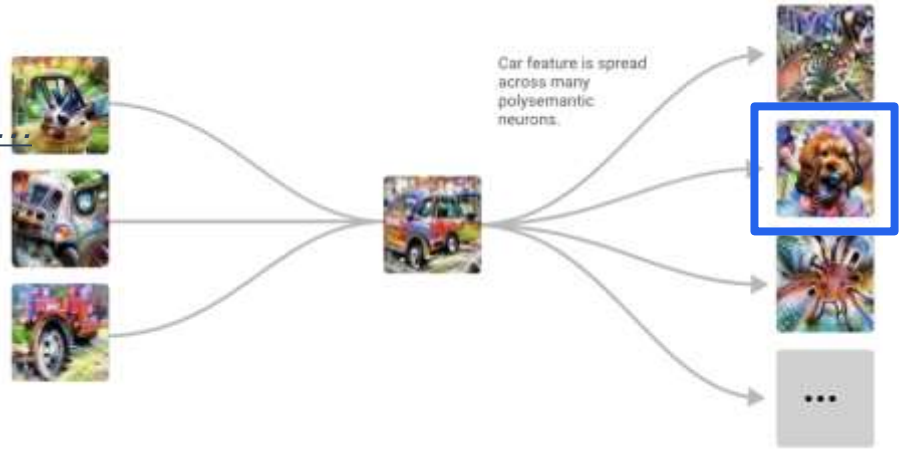
1. Revisiting Canonical Evidence

Circuit → Dog-and-Car Circuit

- But then the model does something surprising. Rather than create another pure car detector at the next layer, it spreads its car feature over a number of neurons that seem to primarily be doing something else — in particular, dog detectors.

"This circuit suggests that polysemantic neurons are, in some sense, deliberate. …… detecting cars and dogs was deeply intertwined … … had a "pure neuron" and then mixed it up with other features.

We call this phenomenon superposition."



https://distill.pub/2020/circuits/zoom-in/

1. Revisiting canonical evidence

The Dominant View in Prior Interpretability:



- ***Polysemantic*** neurons are common
- Neural networks rely on ***Superposition***
- Neurons are not the right unit of analysis
- Interpretability requires first "untangling" the network

2. Neural Networks Are Already Clean

Nick argues that much of the canonical evidence for polysemanticity has been misinterpreted:

- Feature Visualization → The most activating image does not necessarily reveal the neuron's true representation.
- Regularization → Hand-crafted priors (e.g., blur, jitter, and natural image priors) bias the optimization toward human-recognizable natural objects.
- Human interpretation → We tend to interpret neurons through object-centric concepts rather than underlying patterns or functions.

2. Neural Networks Are Already Clean

Validating Superposition: Shared Representations Don't Disappear with Scale



- Models with different scales:

  ConvNeXt-Small/Medium/Large, DINOv2

- Hypo (Superposition): More capacity → disentangled, object-specific pose neurons

- Conclusion: More capacity leads to richer and more detailed representations, not necessarily more disentangled ones -- animals and cars may still be represented together.

2. Neural Networks Are Already Clean



Nick further corrected:

- Cat-and-Car neuron → V-shape detector
- Car-to-Dog circuit → Shared pose representation



The same neuron shows up in DINOv2 and in our large and medium ConvNeXts: dogs and vehicles grouped by the way they face the camera. Our smallest model only has a rough version, with cars and dogs together but not by pose.

How an object faces the camera:

(i) the smallest, only a coarse shared visual pattern; (ii) While larger models organize more precisely by pose

2. Neural Networks Are Already Clean



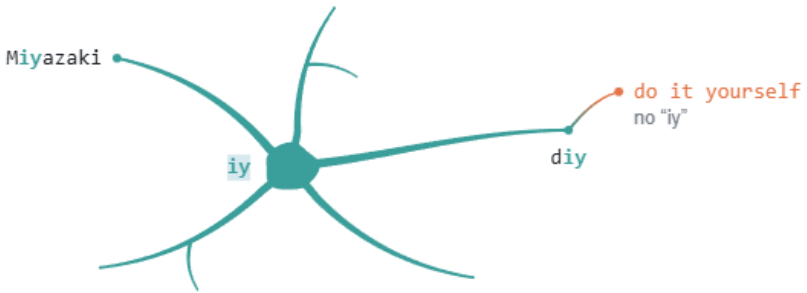
Nick summarized:

What looked like superposition may simply be ordinary ***Generalization***.

2. Neural Networks Are Already Clean

Recursive Generalization:

- A neuron generalizing along more than one axis, often first along one axis, then sprouting outwards in ways that are easy to follow locally, but stop it from being “monosemantic” in the end.



Miyazaki (宮崎) → iy

→ diy → do it yourself (w/o iy)

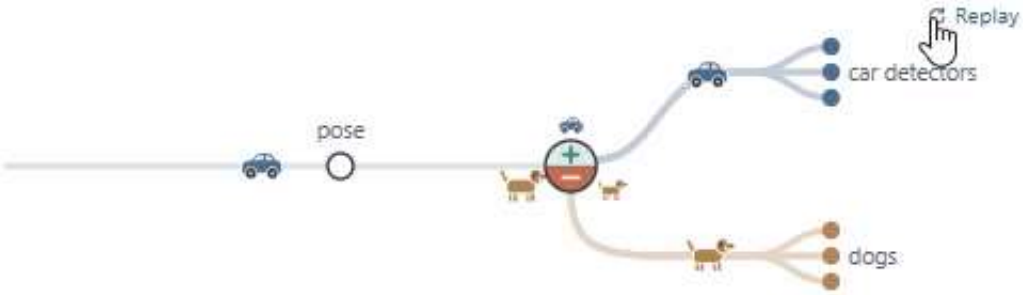
Recursive generalization. An “iy” neuron grows in several directions. From “diy” it reaches “do it yourself,” which has no “iy” at all. Conceptual illustration.

Replay ↻

2. Neural Networks Are Already Clean

Nick pointed out:

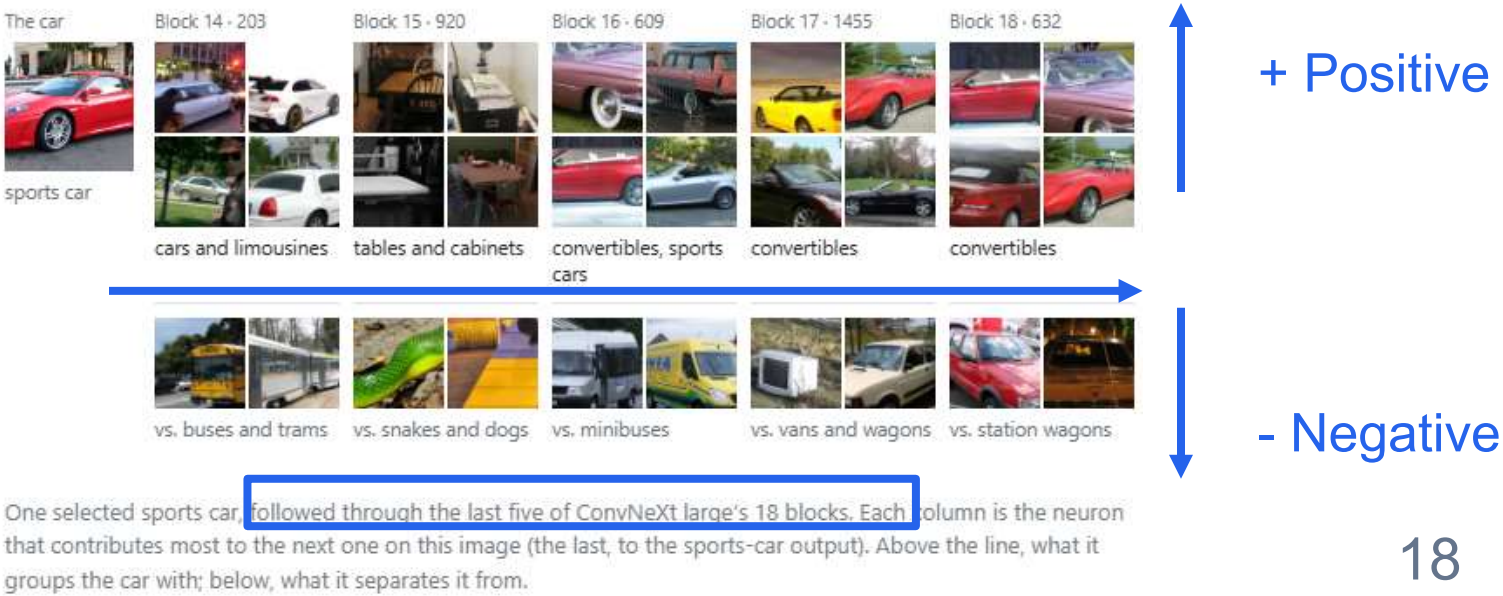
- ***Neurons more as control flow.*** A neuron is less about what it sees than about what it does to the flow coming in, and often its job is to gate a region of the network.



Cars and dogs share a pose early on; a gate in front of the car detectors lets cars through and turns dogs away with its negative side.

2. Neural Networks Are Already Clean

In this way, the negative side of neurons matters almost as much as positives for understanding them



One selected sports car, followed through the last five of ConvNeXt large's 18 blocks. Each column is the neuron that contributes most to the next one on this image (the last, to the sports-car output). Above the line, what it groups the car with; below, what it separates it from.

2. Neural Networks Are Already Clean

Even in the smallest model, the positive and negative sides remain differentiated, though more coarsely, and this differentiation ends earlier.



The small model's last-layer neuron: ground vehicles on its positive side, boats and planes on its negative side.

3. Discussion and Open Questions

What Can We Learn from This?

1. As the author puts it, "*The more we read it the way its own readers do, the more interpretable I think it will look. We should be wary of reading it too much like a human, and concluding it's messy just because it doesn't fit our preconceived concepts.*"

What appears uninterpretable may reflect the limits of our human-centered perspective rather than genuine disorder in the model?

3. Discussion and Open Questions

What Can We Learn from This?

2. As the author puts it, "*The solution might look like a laboratory run by automated scientists who watch the network through training and inference ····· Capable agents were built for other work, and modern architectures were chosen because they learn well. Both happen to serve interpretability.*"

Can we trust one black-box model to interpret another, or does this merely shift the interpretability problem?